

Philip Woods

CURRICULUM VITAE

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Work

NORTHWESTERN UNIVERSITY CENTER FOR CONNECTED LEARNING AND COMPUTER-BASED MODELING
Research assistant 2017 – 2018
Advisors: Uri Wilensky and Hillary Swanson

Education

CALIFORNIA INSTITUTE OF TECHNOLOGY, PASADENA, CA
Ph.D. candidate, Geobiology 2018 – present
Advisor: Victoria Orphan
Committee: Victoria Orphan, Jared Leadbetter, Woody Fischer, Joe Parker

HARVEY MUDD COLLEGE, CLAREMONT, CA
B.S., Joint Major in Chemistry and Biology May 2017
Graduated with Distinction
Thesis: *The effects of ribosomal DNA copy number on lifespan, health, and dietary restriction.*
Advisor: Jae Hur

Publications

1. Shalvarjian, K. E., Chadwick, G. L., Pérez, P. I., **Woods, P. H.**, Orphan, V. J. & Nayak, D. D. Methanogenic archaea encoding Pyrrolysine maintain ambiguous amber codon usage. *Proceedings of the National Academy of Sciences* **122**, e2517473122. <https://www.pnas.org/doi/10.1073/pnas.2517473122> (2025).
2. **Woods, P. H.**, Speth, D. R., Laso-Pérez, R., Utter, D. R., Ruff, S. E. & Orphan, V. J. Identification of key steps in the evolution of anaerobic methanotrophy in Candidatus Methanovorans (ANME-3) archaea. *Science Advances* **11**, eadq5232. <https://www.science.org/doi/full/10.1126/sciadv.adq5232> (2025).
3. Arastoopour Irgens, G., Dabholkar, S., Bain, C., **Woods, P.**, Hall, K., Swanson, H., Horn, M. & Wilensky, U. Modeling and Measuring High School Students' Computational Thinking Practices in Science. *Journal of Science Education and Technology* **29**, 137–161. <https://doi.org/10.1007/s10956-020-09811-1> (2020).

Presentations

CONTRIBUTED TALKS

1. **Woods, P.** *Convergence and horizontal gene transfer drive evolution of anaerobic methanotrophy in archaea.* Southern California Evolutionary Genetics and Genomics Meeting. Pasadena, CA. (2024).
2. **Woods, P.** *Convergence and horizontal transfer drive the evolution of anaerobic methanotrophy in ANME-3.* Southern California Geobiology Symposium. Santa Barbara, CA. (2023).

CONTRIBUTED POSTERS

1. **Woods, P.** *Investigating three-dimensional spatial organization in the interdomain syntrophy between ANME archaea and sulfate-reducing bacteria using FIB-SEM.* AbSciCon. Madison, WI. (2026).
2. **Woods, P.** *Investigation of "Candidatus Methanovorans" reveals early steps in the evolution and ecology of anaerobic methanotrophy.* Geobiology GRC. Ventura, CA. (2026).

3. **Woods, P.** *Convergence and horizontal gene transfer drive evolution of anaerobic methanotrophy in archaea.* Center for Evolutionary Science Symposium. Pasadena, CA. (2025).
4. **Woods, P.** *Toward cultivation and ecological characterization of ANME-3 from an in situ colonization experiment.* New Lineages of Life. Las Vegas, NV. (2025).
5. **Woods, P.** *Using evolution to investigate key features of MCR in methanogens and anaerobic methanotrophs.* Southern California Geobiology Symposium. Pasadena, CA. (2024).
6. **Woods, P.** & Mayr, M. J. *Systems Level Insights Into The Physiology Of Methane Fueled Syntrophy Between ANME Archaea And Sulfate-Reducing Bacteria.* DOE GSP and ECR Annual Meeting. North Bethesda, MD. (2024).
7. **Woods, P.** *Convergence and horizontal transfer drive evolution of anaerobic methanotrophy.* SCASM 86th Annual Meeting. La Jolla, CA. (2022).

INVITED SEMINARS

1. **Woods, P.** *Identification of key steps in the evolution of anaerobic methanotrophy in ANME-3 archaea.* EVOxZankou. Pasadena, CA. (2025).

OTHER SUBMISSIONS

1. Swanson, H., Irgens, G. A., Bain, C., Hall, K., **Woods, P.**, Rogge, C., Horn, M. & Wilensky, U. *Characterizing computational thinking in high school science* in *Proceedings of International Conference of the Learning Sciences, ICLS 13th International Conference of the Learning Sciences, ICLS 2018: Rethinking Learning in the Digital Age: Making the Learning Sciences Count.* **2** (International Society of the Learning Sciences, 2018), 871–878. <https://www.scholars.northwestern.edu/en/publications/characterizing-computational-thinking-in-high-school-science>.

Fellowships, Honors, and Awards

Certificate of Achievement, <i>American Society for Microbiology, Southern California branch</i>	2022
Benjamin M. Rosen Graduate Fellowship, <i>California Institute of Technology</i>	2018
CEMI 1st Year Fellowship, <i>California Institute of Technology</i>	2018
Mindlin Prize for Innovative Ideas in the Sciences, <i>Harvey Mudd College</i>	2017
Best Senior Thesis in Biology, <i>Harvey Mudd College</i>	2017
Departmental Honors in Biology, <i>Harvey Mudd College</i>	2017

Grants

NASA Astrobiology Program, AbSciCon 2026 Travel Support, \$1000	2026
Caltech Center for Environmental Microbial Interactions, Geobiology Gordon Research Conference and Gordon Research Seminar Travel Support, \$2000	2025
Caltech Center for Environmental Microbial Interactions, “Targeted sequencing of ANME-3 genomes from enrichment cultures and FACS”, \$7,500	2022 – 2023

Research Experience

CALIFORNIA INSTITUTE OF TECHNOLOGY

Advisor: Victoria Orphan 2019 – present

- Maintained long-term enrichment cultures of methane-oxidizing microbial communities.
- Performed FISH and FIB-SEM imaging of ANME samples.
- Used comparative genomics to identify distinctive genome features of the ANME-3 clade.
- Identified several key evolutionary processes in the evolution of ANME from methanogens.

CALIFORNIA INSTITUTE OF TECHNOLOGY

Advisor: Jared Leadbetter 2019 – 2021

- Investigated the manganese-dependent growth potential of a novel *Methylobacterium* isolate.
- Established growth yields and growth kinetics for the *Methylobacterium* under several conditions.
- Assayed manganese oxidation in microbial cultures using spectrophotometry.

NORTHWESTERN UNIVERSITY CENTER FOR CONNECTED LEARNING AND COMPUTER-BASED MODELING
Advisors: Uri Wilensky and Hillary Swanson 2017 – 2018

- Developed multiple NetLogo computational models of CRISPR-Cas in bacterial cells and communities.
- Developed a NetLogo computational model of the evolution of anisogamy.
- Designed experiments using this model to address questions in theoretical biology literature.
- Developed, implemented, and assessed high school science curricula to teach computational thinking.

HARVEY MUDD COLLEGE BIOLOGY DEPARTMENT

Advisor: Jae Hur 2016 – 2017

- Designed and executed a research agenda independently as a senior thesis project.
- Investigated the molecular mechanisms behind dietary restriction-based lifespan extension.
- Performed lifespan, dietary restriction, stress, and qPCR assays on *Drosophila*.

POMONA COLLEGE BIOLOGY DEPARTMENT

Advisor: Melissa Petreaca 2014

- Studied the role of the LIGHT protein in mouse neutrophil degranulation and NETosis.
- Analyzed NET formation using ELISA and fluorescence microscopy.

HARVEY MUDD COLLEGE CHEMISTRY DEPARTMENT

Advisor: Katherine van Heuvelen 2013

- Synthesized cyclam derived ligands for nickel complexes to model cofactor F430.
- Characterized these with Gaussian DFT calculations and UV-vis, NMR, and IR spectroscopy.
- Tested the model complexes for methane oxidation potential using spectrophotometry.

Teaching Experience

CALIFORNIA INSTITUTE OF TECHNOLOGY

GE/ESE/BI 178: MICROBIAL ECOLOGY Winter 2023, Winter 2025

Instructors: Victoria Orphan

- Developed assignments covering physical constraints, taxonomy, and evolution of microbial communities.
- Organized and ran regular discussion of relevant research literature with students.

BI/GE/ESE 105: EVOLUTION

Winter 2020, Winter 2024

Instructors: Victoria Orphan and Rob Phillips

- Developed and implemented lessons covering phylogeny, population genetics, speciation, and coevolution.
- Helped facilitate weekly laboratory exercises on concepts and techniques relevant to evolution.

Service & Outreach

REVIEWER

- Frontiers in Microbiology

CALIFORNIA INSTITUTE OF TECHNOLOGY

Coordinator of the Geobiology Reading Group 2025 – 2026

- Led weekly discussions of recently published articles in the field of geobiology.

Co-coordinator of the Southern California Geobiology Symposium 2024

- Organized conference venue, agenda, talks, and posters.

Co-founder & treasurer of the student chapter of the American Society for Microbiology 2022 – 2023

- Coordinated seminars with invited speakers from domestic and international institutions.

Professional Associations

American Society for Microbiology, Member 2022 – present

Phi Lambda Upsilon, Member 2017 – present

Technical Skills

COMPUTATIONAL SKILLS

- Proficient with the Anvi'o metagenomic tool suite
- Experience with genome assembly and binning
- Proficient with 16S metabarcoding and community composition analysis
- Proficient with Python, Bash scripting, and NetLogo
- Experience with R, Matlab, Javascript, C++, Java, Igor Pro, and Racket.
- Proficient in building agent-based models

INCUBATIONS AND CULTURING

- Proficient in anaerobic culture maintenance
- Experience using anaerobic glove boxes

MICROSCOPY

- Proficient with fluorescence in-situ hybridization
- Experience with BONCAT labeling
- Experience with three-dimensional feature segmentation